

Internship Report

Title: Data Analytics Internship – Google Play Store App Analysis using Power BI & Python

Name: Pratham H Shetty

Domain: Data Analytics

Tools Used: Power BI, Python, Jupyter Notebook, Pandas

Duration: 17th FEB to 17th MAR

1. Introduction

This internship was undertaken with the objective of developing strong practical skills in data analytics, business intelligence, and data visualization using industry-standard tools such as Power BI and Python. The project was based on the Google Play Store dataset, where multiple business-driven analytical tasks were performed.

Through this internship, I gained hands-on experience in key skills such as **data cleaning**, **KPI development**, and **dashboard design**, helping me understand how raw data can be converted into meaningful business insights.

2. Background

In today's data-driven world, organizations rely heavily on analytics to make informed decisions. The Google Play Store contains millions of applications with diverse characteristics such as ratings, reviews, installs, size, category, and revenue models. Analyzing such data provides valuable insights into market trends, user behavior, and application performance.

This internship focused on solving real-world analytical tasks using structured datasets and converting them into interactive dashboards and visual reports, simulating professional industry projects.

3. Learning Objectives

The primary learning objectives of this internship were:

- To understand data cleaning and preprocessing techniques
- To build interactive dashboards using Power BI
- To implement advanced business logic using DAX
- To perform data analysis using Python & Jupyter Notebook
- To develop visual storytelling skills

- To understand time-based filtering and dynamic visualization logic
 - To gain exposure to real-world analytics problem solving
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4. Activities and Tasks

During the internship, I completed six major analytical tasks, each focusing on different visualization techniques and business problems.

Task 1 — Data Cleaning & Exploratory Analysis

- Cleaned dataset
- Converted data types
- Handled missing values
- Performed descriptive analysis

Task 2 — Choropleth Map Analysis

- Created a world map showing installs by category
- Applied category and install-based filters
- Implemented time-based visualization logic

Task 3 — Dual Axis Visualization

- Compared Average Installs vs Average Revenue
- Implemented Free vs Paid app comparison
- Used advanced DAX measures
- Applied business rules and time-based visibility

Task 4 — Time Series Trend Analysis

- Built trend line charts
- Highlighted growth periods >20%
- Applied category translations
- Implemented strict business filters

Task 5 — Bubble Chart Analytics

- Built bubble charts for size vs rating
- Used installs as bubble size
- Applied sentiment analysis preprocessing
- Implemented category translation & time-based display logic

Task 6 — Stacked Area Chart Analytics

- Built cumulative installs over time visualization
- Applied advanced filtering logic
- Implemented category translation
- Added growth highlighting (>25%)
- Implemented dynamic time window logic

Additionally, all tasks were replicated using Python and Jupyter Notebook for enhanced analytical understanding.

5. Skills and Competencies Gained

During this internship, I developed the following skills:

Technical Skills

- Power BI Dashboard Development
- DAX Calculations
- Data Modeling
- Data Cleaning using Power Query
- Python Programming
- Pandas Data Analysis
- Plotly & Matplotlib Visualization

Analytical Skills

- Data interpretation
- Business logic application
- Pattern detection
- Trend analysis

Professional Skills

- Problem-solving
 - Logical thinking
 - Analytical reporting
 - Documentation
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6. Feedback and Evidence

Throughout the internship, each task was implemented as per detailed instructions and validated using:

- Power BI interactive dashboards
- Jupyter Notebook analytical workflows
- Proper documentation
- GitHub repository for project storage

Screenshots, dashboards, Python notebooks, and GitHub repositories serve as evidence of task completion and technical implementation.

7. Challenges and Solutions

Challenge 1 — Dirty Dataset

- Problem: Text values in numeric columns, missing values
- Solution: Applied data cleaning techniques in **Power Query & Pandas**

Challenge 2 — Complex Business Filters

- Problem: Multiple filtering conditions
- Solution: Created stepwise filtering pipelines

Challenge 3 — DAX Time-Based Logic

- Problem: Implementing **time-based visualization visibility**
- Solution: Created **dynamic DAX** measures using **system time functions**

Challenge 4 — Sentiment Analysis Column

- Problem: Dataset lacked text reviews
 - Solution: Designed **Python preprocessing workflow** and implemented workaround logic
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8. Outcomes and Impact

- Developed **6 complete business dashboards**
- Built **Python analytical notebooks**
- Improved understanding of **real-world data analytics workflows**
- Gained strong confidence in **Power BI & Python integration**
- Created **portfolio-grade projects** for career development

This internship significantly enhanced my **technical confidence, analytical ability, and professional readiness** for data analytics roles.

9. Conclusion

This internship provided a valuable learning experience in data analytics and visualization. By working on real datasets and complex analytical tasks, I gained hands-on exposure to business intelligence tools, advanced analytics, and dashboard storytelling techniques.

The internship strengthened my interest in data analytics and has equipped me with practical industry-relevant skills, preparing me for future roles in data science, analytics, and business intelligence.